



Minor Project (21CSA697A)

Proposal (Individual Mode)

Title: Deep Learning-Based Music Genre Classification Using Mel-Spectrograms and CNN

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Abstract :

This project proposes the development of a Deep Learning-based Music Genre Classification system that automatically identifies the genre of a given audio clip using Mel-Spectrogram image analysis.

Traditional music genre classification methods rely on manually engineered audio features, which often fail to capture the complex patterns present in music. This project addresses that limitation by converting raw audio signals into Mel-Spectrogram images using the Librosa library, and then training Convolutional Neural Network (CNN) architectures to classify genres directly from these visual representations of sound.

The system will be trained and evaluated on the GTZAN Dataset, which contains 1,000 audio clips spanning 10 music genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. A baseline custom CNN model will first be implemented, followed by Transfer Learning using pre-trained architectures such as VGG16 or ResNet50 to improve classification accuracy.

The final system will be deployed as an interactive web application using Streamlit, allowing users to upload an audio file and receive real-time genre predictions. Model performance will be evaluated using accuracy, precision, recall, F1-score, and confusion matrix analysis. This project demonstrates a practical, software-only application of deep learning in Music Information Retrieval (MIR).

Assumptions / Declarations :

Assumptions:

1. The GTZAN dataset available on Kaggle is sufficient for training and evaluating the model within the minor project scope.
2. A standard laptop or Google Colab (free tier) will provide adequate computational resources for training the CNN model.
3. Audio clips are preprocessed and noise-free as provided in the GTZAN dataset.
4. The 30-second audio clips in the dataset are representative of each genre.

Declarations:

1. This project is entirely software-based and does not require any hardware components.
2. All tools, libraries, and datasets used are freely available and open-source.

3. This is an individual project developed as part of Minor Project course (21CSA697A).
4. The work is original and will not involve plagiarism of any existing implementation.

Main Objective / Deliverable :

Main Objective:

To build an accurate and deployable Music Genre Classification system using Deep Learning techniques that classifies audio clips into one of 10 genres by analyzing Mel-Spectrogram images through Convolutional Neural Networks.

Key Deliverables:

1. A trained CNN model capable of classifying music into 10 genres with high accuracy.
2. A comparative evaluation of a custom CNN model versus Transfer Learning (VGG16 / ResNet50).
3. A Mel-Spectrogram visualization pipeline using Librosa for audio feature extraction.
4. A Streamlit web application where users can upload audio and get genre predictions.
5. A final project report documenting methodology, experiments, and results.

Timeline and Milestones :

Sl.	Milestones	Timeline
1.	Literature Review and Dataset Collection (GTZAN)	Week 1 – Week 2
2.	Data Preprocessing and Mel-Spectrogram Generation using Librosa	Week 3 – Week 4
3.	CNN Model Design and Training on Spectrogram Images	Week 5 – Week 7
4.	Transfer Learning with VGG16 / ResNet – Fine-tuning	Week 8 – Week 9
5.	Model Evaluation and Performance Analysis (Accuracy, F1-Score)	Week 10
6.	Streamlit Web Application Development and Deployment	Week 11
7.	Testing, Bug Fixes and Final Report Preparation	Week 12

Tools to be Used :

Software / Hardware Tools	Specifications / Purpose
Python 3.10+	Primary programming language
TensorFlow / Keras	Deep learning model building and training
Librosa	Audio processing and Mel-Spectrogram generation
OpenCV	Image preprocessing of spectrogram images

NumPy / Pandas	Numerical computations and data handling
Matplotlib / Seaborn	Visualization of results and training metrics
Scikit-learn	Model evaluation (accuracy, F1, confusion matrix)
Streamlit	Web application for live music genre prediction
Jupyter Notebook	Experimentation and documentation
GTZAN Dataset	10 music genres, 100 audio clips each (30 seconds)

Learning Involved :

Topic	Description
Audio Signal Processing	Understanding audio data, sampling rates, and feature extraction using Librosa
Mel-Spectrogram Generation	Converting audio waveforms into 2D image representations for CNN input
Convolutional Neural Networks	Building and training CNN architectures for image classification tasks
Transfer Learning	Fine-tuning pre-trained models (VGG16 / ResNet50) for music classification
Model Evaluation	Using accuracy, F1-score, precision, recall, and confusion matrix for performance analysis
Web Deployment	Creating an interactive music genre prediction app using Streamlit

Date: 31/05/2026

Student Name and Signature:

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